

Dr. Divyaratan Popli

Bioinformatician Ph.D. (Bioinformatics)

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- Bioinformatician with 8+ years of experience developing computational methods for genomics, with publications in *Nature*, *PNAS*, and *Genome Biology*.
- First-authored software tools for genetic relatedness estimation and population structure analysis, widely applied in archaeogenetics research.
- Working on cancer genomics and personalized medicine, leading multi-omics analyses of rare cancer cohorts integrating WES, WGS, and RNA-seq data for clinical variant discovery.

Skills

- Cancer genetics and personalized medicine
- Programming languages: Python, R, C++, C and Linux shell scripting
- Bayesian statistics, probabilistic modeling
- Workflow management (snakemake)
- Human population genetics and GWAS
- Algorithms and software development
- Multi-omics analysis (genomics, transcriptome)
- Data visualization and dimensionality reduction
- Machine Learning
- Biological networks analysis

Experience

November, 2024 - present

Bioinformatician

NCT Berlin - German Cancer Research Center (DKFZ)

- Currently leading multi-omics analyses of two rare cancer cohorts, integrating WES, WGS, and RNA-seq data to identify clinically relevant variants and molecular signatures, and preparing the results for two peer-reviewed publications.
- Designed and implemented data management workflows for 700+ genomic samples within the SODAR and cubi-tk ecosystem as part of the DKTK master project. Developed custom scripts to identify discrepancies between HPC storage and SODAR repositories and automated data synchronization while preserving complex metadata (e.g., multiple longitudinal DNA/RNA samples per patient).
- Executed and benchmarked the neoantigen prediction pipeline NeoDisc using Singularity, to compare somatic variant detection and predicted neoepitopes against pVACtools.
- Conducted comparative variant analysis on WES data from FFPE and fresh frozen samples, evaluating differences in SNV and indel calling using the in-house Snappy pipeline, and delivered evidence-based recommendations to clinicians supporting the preferential use of fresh frozen preservation.

- Performed cross-platform benchmarking between Snappy and Illumina DRAGEN, assessing concordance in somatic variant calls and pipeline performance using WGS datasets.

April, 2024 - September, 2024

Postdoctoral Researcher

Max Planck Institute for Evolutionary Anthropology, Leipzig

- Designed bioinformatic pipelines to perform quality control for large ancient DNA datasets, and to implement population genetic analyses to understand the relationship of the samples to each other and to modern-day samples.
- Developed a user-friendly snakemake pipeline to implement joint estimation of probabilistic PCA and F-statistics.

October, 2018 - March, 2024

PhD candidate

Max Planck Institute for Evolutionary Anthropology, Leipzig

- Designed an algorithm to effectively estimate genetic relatedness from ancient DNA datasets, while modeling the effects of DNA contamination and inbreeding. This method is available as a python package on Github in the repository Kinship Inference.
- Developed a statistical framework to jointly perform probabilistic PCA with F-statistics to quantify the PCA results. This pipeline is available on Github as a snakemake pipeline.
- Contributed to collaborative projects resulting in high-impact publications, including genetic insights into Neanderthal social organization (*Nature*, 2022) and social practices of Avar communities (*Nature*, 2024).

July, 2014 - June, 2016

Research Fellow

Indian Institute of Science, Bangalore

- Designed electrophysiology experiments to study the effect of ultrasound on cultured neuronal networks.
- Applied Graph Theory and statistical tests to analyze the results.

April, 2013 - August, 2013

Research Assistant

University of Amsterdam

- Designed cognitive tasks to study identification of Task-Unrelated Thoughts (TUTs), and implemented these tasks on a stimulus delivery and experiment control program for neuroscience.
- Conducted experiments with over 34 subjects.
- Analyzed the measured pupil diameter data to show correlation with incidence of TUTs.

November, 2012 - February, 2013

Research Assistant

National Brain Research Center, Manesar

- Automated the search for miRNA molecules that target mRNA in neurons from online databases.
- Analyzed the specificity of miRNAs that are known to function at synapses to control dendritic spine structure.

Academic Qualifications

October, 2018 - March, 2024

Ph.D. (Bioinformatics)

Thesis Title: Estimating relatedness with ancient DNA

Max Planck Institute for Evolutionary Anthropology, Leipzig

affiliated to Faculty of Mathematics and Informatics, Leipzig University

July, 2016 - June, 2017

Master of Science (Biology)

Thesis Title: Effect of Ultrasound on Neuronal Network Communication

Indian Institute of Science, Bangalore

April, 2010 - June, 2014

Bachelor of Technology (Biological Sciences and Bioengineering)

Indian Institute of Technology, Kanpur

Teaching experience

- Delivered lectures for master's students in the course Molecular Anthropology (Leipzig University, 2021–2023), covering genetic diversity, natural selection, and molecular mechanisms of recombination and mutation.
- Taught a one-year physics course (2017–2018) to high school students, focusing on fundamental concepts and quantitative problem-solving for competitive entrance examinations (e.g., IIT-JEE).
- Available to teach courses on bioinformatics, population genetics and ancient DNA, cancer genetics, and applied statistics.

Awards and Scholarships

- **GATE 2017 (Ecology & Evolution)**, All India Rank 6
- **GATE Fellowship** (2014–2017)

- **IIT-JEE 2010**, All India Rank **2014**
- **Merit-cum-Means (MCM) Scholarship**, Indian Institute of Technology Kanpur (2010–2014)
- **IMPRS Scholarship**, International Max Planck Research School for Evolutionary Anthropology, Leipzig (2018–2024)

Publications

- Massilani, D., Peyrégne, S., Iasi, L. N. M., ..., **Popli, D.**, ..., Peter, B. M., Kelso, J., & Pääbo, S. (2026). *A high-coverage Neandertal genome from the Altai Mountains reveals population structure among Neandertals*. *Proceedings of the National Academy of Sciences*, 123(13), e2534576123.
- Isacchini, G., Okada, T., Schmid, C., **Popli, D.**, Peter, B. M., Schiffels, S., & Hallatschek, O. (2026). *Inferring Migration Networks with Time-Lagged F2 Statistics*. *bioRxiv* (preprint).
- Rohrlach, A. B., Tuke, J., **Popli, D.**, & Haak, W. (2025). *BREADR: An R package for the Bayesian estimation of genetic relatedness from low-coverage genotype data*. *Journal of Open Source Software*, 10(108), 7916.
- Gneccchi Ruscone, G. A., Rácz, Z., Samu, L., ..., **Popli, D.**, ..., Hofmanová, Z. (2024). *Network of large pedigrees reveals social practices of Avar communities*. *Nature*, 629, 376–383.
- **Popli, D.** & Peter, B. M. (2024). *A joint framework for studying population structure using PCA and F-statistics*. *bioRxiv* (preprint).
- **Popli, D.**, Peyrégne, S., & Peter, B. M. (2023). *KIN: A method to infer relatedness from low-coverage ancient DNA*. *Genome Biology*, 24 (10).
- Skov, L., Peyrégne, S., **Popli, D.**, Iasi, L. N. M., ..., Peter, B. M. (2022). *Genetic insights into the social organization of Neanderthals*. *Nature*, 610, 519–525.

Conference Proceedings

- **Popli, D.**, Peyrégne, S., & Peter, B. M. *Estimating genetic relatedness from ancient DNA*. In: Meller H, Krause J, Haak W, Risch R, eds. *Kinship, Sex, and Biological Relatedness: The Contribution of Archaeogenetics to the Understanding of Social and Biological Relations*. 15th Central German Archaeologists' Day, Halle (Saale), 2022. Propylaeum; 2023.